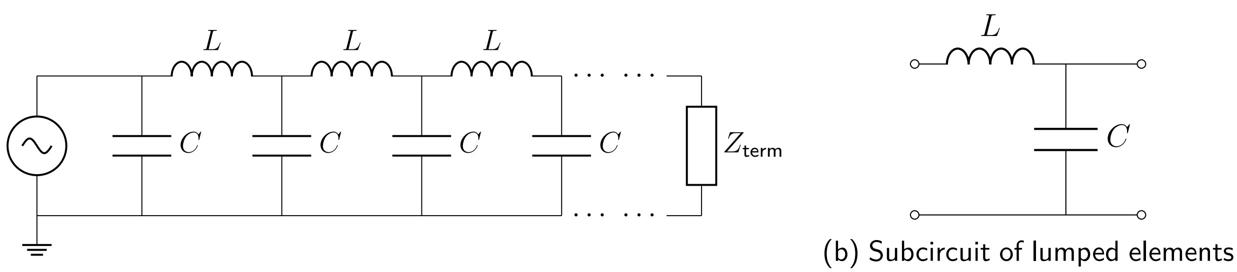
Lab 3: Speed of Light

Objective

The objective of this lab is to measure the speed of light in a coaxial and twin-lead waveguide as well as to measure the wave's attenuation and reflection characteristics from an impedance at the end of the waveguide.

Theory

An ideal lossless waveguide has an inductance per unit length, L', and a capactiance per unit length, C', obtained from solving the electric and magnetic fields for the geometry of the waveguide. For coaxial and twin-lead lines, the fields support a TEM mode, allowing the use of distributed circuit parameters L' and C'. We can imagine the whole waveguide as being built up from infinitesimal inductance and capactiance elements for each dx of length of the waveguide, as illustrated in Figure 1a. The waveguide may be terminated with an impedance, Z_{term}, which is ∞ for an open-termination, 0 for a shorted-termination, or any other value we choose. The choice of Z_{term} will affect the reflection of waves from the end of the waveguide.



(a) Waveguide composed of lumped elements

(b) Subcircuit of lumped elements in dx

Figure 1: Lossless waveguide composed of ideal inductors and capacitors considered as lumped elements.

Telegrapher's Equations

Consider treating the transmission line as a lossless distributed circuit: instead of one lumped inductor and one lumped capacitor, each small length dx of line has

$$L = L'\,dx \quad \text{(series inductance)}$$

and

$$C = C'\,dx \quad \text{(shunt capacitance)},$$

as illustrated in Figure 1. From this model we can derive the *telegrapher's equations*, and from those we get both the wave speed and the characteristic impedance of the transmissionline / waveguide.

Take a small segment of transmission line of length dx. Let

- $V(x,t) =$ voltage between the conductors at position x,
- $I(x,t) =$ current along the line at position x.

Apply Kirchhoff's voltage law to the small segment, Figure 1b. The voltage drop across the series inductance is

$$dV = L' dx \, \frac{\partial I}{\partial t}.$$

Because voltage decreases in the direction of increasing x,

$$V(x,t) - V(x+dx,t) = L' dx \, \frac{\partial I}{\partial t}.$$

So

$$-\frac{\partial V}{\partial x} dx = L' dx \, \frac{\partial I}{\partial t},$$

hence

$$\boxed{\frac{\partial V}{\partial x} = -L' \frac{\partial I}{\partial t}}.$$

Now apply Kirchhoff's current law. The current difference between the left and right ends of the segment must go into charging the shunt capacitor:

$$I(x,t) - I(x+dx,t) = C' dx \, \frac{\partial V}{\partial t}.$$

Thus

$$-\frac{\partial I}{\partial x} dx = C' dx \, \frac{\partial V}{\partial t},$$

so

$$\boxed{\frac{\partial I}{\partial x} = -C' \frac{\partial V}{\partial t}}.$$

The boxed equations are the *lossless* telegrapher's equations.

Wave Equation

The telegrapher's equations can be combined to derive a wave equation for signal propagation on the transmission line / waveguide. To proceed differentiate the voltage equation with respect to x:

$$\frac{\partial^2 V}{\partial x^2} = -L' \frac{\partial}{\partial x} \left(\frac{\partial I}{\partial t} \right).$$

Swap the order of derivatives:

$$\frac{\partial^2 V}{\partial x^2} = -L' \frac{\partial}{\partial t}\left(\frac{\partial I}{\partial x}\right).$$

Now use the current equation,

$$\frac{\partial I}{\partial x} = -C' \frac{\partial V}{\partial t},$$

to get

$$\frac{\partial^2 V}{\partial x^2} = -L' \frac{\partial}{\partial t}\left(-C' \frac{\partial V}{\partial t}\right) = L'C' \frac{\partial^2 V}{\partial t^2}.$$

So

$$\boxed{\frac{\partial^2 V}{\partial x^2} = L'C' \frac{\partial^2 V}{\partial t^2}}.$$

This is a standard wave equation of the form

$$\frac{\partial^2 V}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 V}{\partial t^2},$$

so

$$\boxed{v = \frac{1}{\sqrt{L'C'}}}.$$

The same derivation holds for the current:

$$\boxed{\frac{\partial^2 I}{\partial x^2} = L'C' \frac{\partial^2 I}{\partial t^2}}.$$

So voltage and current both propagate as waves with speed $v = 1/sqrtL'C'$. For TEM lines, one can show from the field solutions that $L'C' = \mu\epsilon$, giving

$$v = \frac{1}{\sqrt{L'C'}} = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\epsilon_r\,\mu_r}},$$

where $c = 1/\sqrt{\mu_0\epsilon_0}$ is the speed of light in vacuum. In non-magnetic dielectrics ($\mu_r = 1$) so that, $v = c/\sqrt{\epsilon_r}$.

Characteristic Impedance, Z_0

Now assume a forward-traveling wave:

$$V(x,t) = V_0\, f(x - vt), \qquad I(x,t) = I_0\, f(x - vt).$$

For such a wave,

$$\frac{\partial f}{\partial x} = f', \qquad \frac{\partial f}{\partial t} = -vf'.$$

Substitute into

$$\frac{\partial V}{\partial x} = -L' \frac{\partial I}{\partial t}.$$

That gives

$$V_0 f' = -L'(-vI_0 f') = L' v I_0 f'.$$

Assuming $f' \neq 0$,

$$V_0 = L' v I_0.$$

Therefore

$$\frac{V_0}{I_0} = L' v.$$

Using

$$v = \frac{1}{\sqrt{L'C'}},$$

we get

$$\frac{V_0}{I_0} = L' \frac{1}{\sqrt{L'C'}} = \sqrt{\frac{L'}{C'}}.$$

So the ratio of voltage to current for a pure traveling wave is

$$\boxed{Z_0 = \frac{V_0}{I_0} = \sqrt{\frac{L'}{C'}}}.$$

Z_0 is the **characteristic impedance** of the transmission line / waveguide. Note: Z_0 is the ratio V/I for a wave traveling in one direction. A load equal to Z_0 absorbs all incident energy, producing no reflection.

Experiment

In this experiment, you will measure v, C', and infer L' and Z_0, then test how impedance matching affects reflections. Note applicable uncertainty in your measurements and do proper error propagation when computing quantities from measured values.

Equipment

You will need the following:

- Function generator
- Digital oscilloscope

- BNC T-connector
- BNC-to-Bananna connector
- Coaxial and twin-lead cable
- Assorted terminations

Waveguide Length

Gently unspool the coax cable being careful not to kink it. Using any method you like determine the length of the cable to the closest centimeter and estimate your uncertainty in the length. Repeat with the twin-lead cable.

Capacitance per unit length, C'

Using the BNC-to-Bananna connector connect the coax cable to the multimeter to measure the capacitance. Note: Ensure the far end of the cable is open (ie $Z_{\text{term}} = \infty$) when measuring capacitance. Estimate the uncertainty in the measured capacitance. Compute the capacitance per unit length of the coax. Repeat for the twin-lead cable.

Speed of Light, Attentuation, and Reflection Coefficients

Use a BNC T-connection to connect the function generator's output to the oscilloscope's channel 1 and the other end of the T-connection to the coax cable. Leave the other end of the coax cable open, (ie $Z_{\text{term}} = \infty$). Set up the function generator to output a 1$V_{\text{p-p}}$ 100kHz square wave with 0.2% duty cycle. Note: The small duty cycle produces short pulses so the incident and reflected signals do not overlap. Set up the oscilloscope to capture a single shot of the square wave and its reflection from the other end of the cable. This may take a few times to get right. Do the following:

(a) Using the cursors on the oscilloscope determine the total round-trip time of flight for the pulse. Estimate the uncertainty in this measurement.

(b) Using the cursors measure the amplitudes of the incident and reflected pulses. What do you notice about the reflected pulse relative to the incident pulse? (in-phase, out-of-phase, etc)

(c) Using your measured values of cable length and total time of flight calculate the speed of light (ie the speed of the electromagnetic pulse) in the coax waveguide with uncertainty. Your value should be less than the speed of light in vacuum but the same power of 10. Note: Your time of flight is the time to go down the cable and come back.

(d) Using your measured amplitudes calculate the round-trip attenuation:

$$\text{Attenuation}_{\text{RT}} = 20 \cdot \log_{10}\left(\left|\frac{V_{\text{refl}}}{V_{\text{inc}}}\right|\right)$$

Divide by twice the cable length to obtain attenuation per unit length for the coax waveguide.

(e) Add the shorted BNC connector (ie $Z_{\text{term}} = 0$) to the end of the coax cable. Note the difference in the reflected pulse and compare to the open-cable result. Record your observations for inclusion in your report.

(f) Repeat these measurements for the twin-lead waveguide.

Inductance per unit length, L', & Line Impedance, Z_0

We have experimentally measured v and C' for each waveguide. Do the following:

(a) Use them to compute L' and the line impedance, Z_0, for each waveguide. What is the relative permitivity, $\epsilon_r = \epsilon/\epsilon_0$, of the dielectric separating the conductors in the coax and twin-lead wave guides (assuming that the permability of the dielectric is the same as μ_0)?

(b) Construct a terminating impedance that equals the characteristic impedance, $Z_{\text{term}} = Z_0$, from the available resistors for each waveguide and attach it to the end of the transmission line / waveguide. Investigate the behavior of pulses reflecting from this terminal impedance for each waveguide.

Report

Follow the IEEE report template for your write up. Take care to account for error propagation in calculated quantites. Submit your report on Blackboard in pdf format.